SESSION 3

Requirement Gathering Report

AI Career Guidance & Skill Recommendation System

Project	AI Career Guidance & Skill Recommendation System
Deliverable	Requirement Gathering Report
Industry Mapping	Business Analysis and Client Interaction
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Repository	[Add GitHub repository link]

Topics Covered in This Session

- Stakeholder identification
- Requirement gathering techniques: interview simulation, questionnaire design, user persona creation
- User story writing

1. Stakeholder Identification

Stakeholder	Type	Interest / Need	Influence
Students	Primary user	Career clarity, skill gap view, roadmap, courses	High
Placement cell / Training officer	Secondary user	Skill trends and gaps across the batch	High
Admin / Content manager	Operator	Maintain careers, skills and courses data	Medium
Faculty and mentors	Indirect	Guide students using insights	Medium
Employers / recruiters	Indirect	Better prepared candidates	Low
Development team	Internal	Clear, stable requirements	High
Claude API provider	External system	Fair usage, valid API key and limits	Medium

Stakeholder map

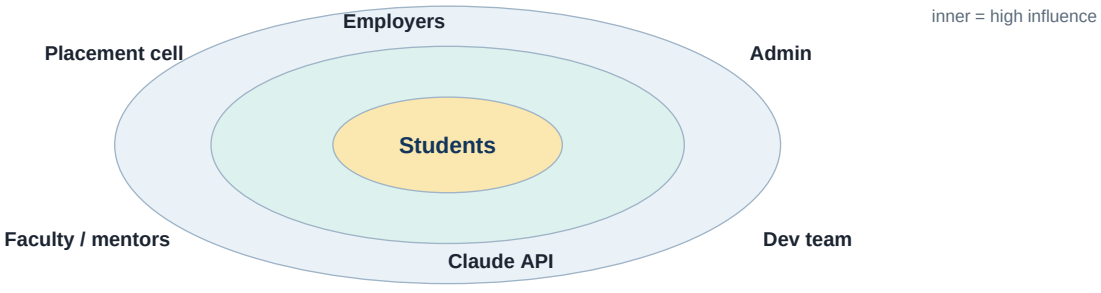

Figure 1: Stakeholder influence map (inner ring = closest to the product)

2. Requirement Gathering Techniques

Technique	How we used it	Output
Interview simulation	Team members role-played a student and a placement officer using a prepared script	Pain points, needs, priorities

Technique	How we used it	Output
Questionnaire	Google Form shared with classmates and juniors	Quantified preferences
User personas	Built from interview and survey patterns	Design targets for UI and features
Document and competitor review	Studied career sites and course platforms	Feature gaps and ideas
Brainstorming	Team sessions to merge ideas	Initial feature list

3. Interview Simulation

3.1 Interview 1: Student (role-played)

Question	Response (simulated)	Requirement insight
How do you currently decide your career path?	"I watch videos and ask seniors. I am still confused between web development and data roles."	Need personalised role comparison
What do you do when you do not know which skills you lack?	"I search job descriptions, but they are all different."	Need automatic skill gap analysis
What would make a roadmap useful?	"Clear order, time estimate and free courses."	Ordered roadmap with duration and course links
Would you upload your resume to a website?	"Yes, if my data stays private."	Privacy and secure storage
Would you ask a chatbot career questions?	"Yes, if it knows my profile."	Profile-aware chatbot

3.2 Interview 2: Placement officer (role-played)

Question	Response (simulated)	Requirement insight
What problem do you face most?	"Many students are unaware of skills that recruiters want."	Common gap analytics
What data would help you?	"Most demanded skills and how many students lack them."	Analytics dashboard
Who updates company skill needs?	"We can, if there is a simple admin panel."	Admin career and course management

4. Questionnaire Design

Target respondents: 2nd to final year engineering students. Suggested sample: 30 or more responses. Responses are anonymous.

No.	Question	Type
1	Which year are you studying in?	Single choice
2	How clear are you about your target career?	Likert 1 to 5
3	Which careers interest you? (Java, Full Stack, Data Analyst, Data Science, DevOps, other)	Multiple choice
4	How do you currently get career guidance?	Multiple choice
5	Do you know which skills you are missing for your target role?	Yes / Partly / No
6	How comfortable are you uploading your resume to a web tool?	Likert 1 to 5
7	Which feature matters most? (recommendation, gap analysis, roadmap, courses, chatbot)	Ranking
8	Preferred learning resources (free videos, paid courses, documentation, projects)	Multiple choice
9	How much time per week can you give to learning a new skill?	Single choice
10	Would you use an AI chatbot for career doubts?	Yes / Maybe / No

No.	Question	Type
11	What worries you about AI-generated career advice?	Open text
12	Any feature you would like us to add?	Open text

5. User Personas

Rahul Verma, 20 3rd-year CSE student, tier-2 college	**Goals:** Choose between Full Stack and Data Analyst and learn the right skills before internships. **Pain points:** Confused by conflicting advice; unsure how strong his profile is. **Behaviour:** Uses mobile and laptop, prefers free resources, learns through projects.
Ananya Singh, 21 Final-year IT student preparing for placements	**Goals:** Find out why she is not getting shortlisted and fix gaps quickly. **Pain points:** Limited time, does not know which missing skill to prioritise. **Behaviour:** Studies in short sessions, wants a clear weekly plan.
Mr. Sharma, 42 Training and Placement Officer	**Goals:** Understand the skill readiness of the whole batch and plan workshops. **Pain points:** Manual surveys, no consolidated data. **Behaviour:** Uses desktop, needs charts and exports, little time for data entry.

6. User Stories

Format: *As a [role], I want [feature], so that [benefit].* Priority uses MoSCoW (Must, Should, Could).

ID	User story	Acceptance criteria	Priority
US-01	As a student, I want to register and log in securely so that my data is private.	Valid email and password create an account; invalid login is rejected; session uses JWT	Must
US-02	As a student, I want to upload my resume so that the system can understand my profile.	PDF/DOCX up to 5 MB accepted; wrong type shows an error	Must
US-03	As a student, I want my skills, education, projects and experience extracted so that I do not type them manually.	Extracted data shown within 15 s; user can edit it	Must
US-04	As a student, I want career recommendations with match scores so that I know which roles suit me.	At least 3 roles ranked by score with reasons	Must
US-05	As a student, I want to see skills I have and skills I lack for a role so that I know what to learn.	Matched and missing skills listed for chosen role	Must
US-06	As a student, I want a personalised roadmap so that I can learn in the right order.	Steps ordered by priority with estimated duration	Must
US-07	As a student, I want course suggestions for each missing skill so that I can start learning immediately.	Each gap shows at least one course with link and level	Should
US-08	As a student, I want to mark roadmap steps complete so that I can track progress.	Progress percentage updates and persists	Should
US-09	As a student, I want to chat with an AI career bot so that I can clear my doubts.	Reply in under 8 s using my profile as context	Should
US-10	As a student, I want to choose a different target career so that I can compare options.	Gap and roadmap regenerate for the new role	Should
US-11	As a student, I want to delete my resume and data so that I control my privacy.	Data removed permanently after confirmation	Should
US-12	As a placement officer, I want an analytics dashboard so that I can see in-demand skills and common gaps.	Charts for top skills, popular careers, common gaps	Should

ID	User story	Acceptance criteria	Priority
US-13	As an admin, I want to add and edit careers and required skills so that recommendations stay current.	Changes reflect in the next recommendation	Must
US-14	As an admin, I want to manage the course catalogue so that suggestions are relevant.	Add, edit, delete courses linked to skills	Should
US-15	As a student, I want to download my roadmap as a PDF so that I can keep it offline.	PDF export with roadmap and courses	Could

7. Requirement Summary

Category	Count	Examples
Functional (from stories)	15	Resume analysis, recommendation, gap analysis, roadmap, chatbot, analytics
Non-functional (derived)	8	Security, privacy, response time, usability, availability
Constraints	4	MERN + Claude API stack, semester timeline, English only, web only

These requirements are formalised in the SRS document (Session 4).